

Mohammad Vali

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Postdoctoral researcher at Aalto University, advised by Prof. Arno Solin, specializing in vision/speech discrete representation learning, with 8+ years of research experience and 15+ peer-reviewed publications (at *ICLR*, *TMLR*, *IEEE TASLP*). Interested in Machine Learning, Computer Vision, Generative AI, Explainable AI, Speech/Audio Processing, and Privacy, seeking for fundamental solutions for research questions.

Experience

- **Postdoctoral Researcher**, Aalto University, Finland *Supervisor: Prof. Arno Solin* (Dec 2024 – Present)
Tasks: (i) Research on improving discrete representation learning in DNNs via vector quantization, and compressing 3D Gaussian splatting, (ii) Co-dvising PhD/MSc/BSc students, (iii) Involved in hiring procedure by shortlisting, interviewing and then advising summer interns, (iv) Teaching assistant for Introduction to AI course.
- **Visitor Researcher**, Inria Research Center, France *Supervisor: Dr. Emmanuel Vincent* (Sep 2024 – Nov 2024)
Topic: Developed speaker anonymization algorithms to conceal speaker's identity while preserving utility in DNN-based downstream tasks, e.g., automatic speech recognition (ASR) and speech emotion recognition (SER).
- **Doctoral Researcher**, Aalto University, Finland *Supervisor: Prof. Tom Bäckström* (Nov 2020 – Nov 2024)
Tasks: (i) Developed novel techniques from scratch that improves codebook learning enabling end-to-end training of DNNs, interpretability in DNNs, and speech privacy, (ii) Co-advising MSc students, (iii) Teaching assistant.

Education

- **Ph.D.**, Aalto University, Finland (2020 – 2024)
GPA: 5.0 / 5.0 | Thesis: Vector Quantization in Deep Neural Networks for Speech and Image Processing ([Link](#))
- **M.Sc.**, Babol Noshirvani University, Iran (2015 – 2017)
GPA: 4.5 / 5.0 | Thesis: Adaptive Image Watermarking
- **B.Sc.**, Babol Noshirvani University, Iran (2009 – 2014)
GPA: 4.0 / 5.0 | Final Project: Simulation and Assembling of the Injection Pump for Electrospinning Machine

Selected Projects

- **Codebook Learning for Vector Quantization in Deep Neural Networks** *Core Researcher* (2022 – Present)
↪ Publications: Vali et al. [2022], Vali et al. [2023], Vali et al. [2025]
Developed **end-to-end training** methods for DNNs that use non-differentiable vector quantization. Evaluated the methods on (i) **image compression** and **speech coding** with **VQ-VAE**, and (ii) **image synthesis** with **VQGAN** that uses **autoregressive transformers**. Results are published in **ICLR 2026**, **IEEE Access** journal, and **ICASSP 2023**.
- **Compressing 3D Gaussian Splatting (3DGS)** *Project Co-Advisor* (2025 – Present)
↪ Publications: Wang et al. [2025].a, Wang et al. [2025].b
Together with Haishan Wang, developed compression methods to reduce the storage cost for **3DGS** that is used for **3D novel view synthesis**. The Gaussian splats' features are compressed using **vector quantization**, **octree-style hierarchy**, **entropy coding**, and **arithmetic coding**. Results are published in **ECCV 2026** and **SCIA 2025**.
- **Interpreting Latent Spaces in Generative Models** *Core Researcher*/Project Lead[‡]* (2023 – 2025)
↪ Publications: Vali et al. [2023]*, Vali et al. [2025]*, Anand et al. [2025][‡]
Developed methods to interpret (i) a **VQ-VAE** used for **voice conversion**, (ii) **StyleGAN2/BigGAN** used for **image generation**, and (iii) to control generations in pre-trained **diffusion models** by **eigenvector decomposition** of **self-attention** weights of **denoising autoencoders**. Published in **TMLR** journal, **ICASSP 2026**, and **Interspeech 2023**.
- **Privacy Preservation in Speech Processing Applications** *Core Researcher*/Collaborator[†]* (2024 – 2025)
↪ Publications: Vali et al. [2024]*, Bäckström et al. [2025][†]
With Bäckström, we showed that bundled side info in speech (e.g., **phone embeddings**) can reveal speaker identity. Inspired by **K-anonymity**, we proposed a generic **information bottleneck** quantization method to filter private speaker-specific info for downstream tasks. Results are published in **IEEE TASLP** journal and **Interspeech 2024**.

Technical Skills

■ Programming

- **Python:** Designed and trained deep neural networks e.g., CNNs, GANs, autoencoders, transformers, diffusion models (**PyTorch**, **TensorFlow**, **Keras**), tracking training logs (**Aim**, **Weights & Biases**, **TensorBoard**), evaluation, visualization (**Matplotlib**, **Altair**, **Seaborn**), data analysis and pre-processing (**Numpy**, **Pandas**).
- **MATLAB**, **C**, **C++**, **R**, **Git**, **CUDA**, **VS Code**, **PyCharm**, **Jupyter**, **Shell scripting**, **LaTeX**.

■ Knowledge

- **Machine Learning:** supervised/unsupervised/self-supervised learning, classification, clustering.
- **Deep Neural Networks:** transformers (LLMs), convolutional neural networks (CNNs), generative adversarial networks (GANs), variational/vector-quantized autoencoders (VQ-VAEs), graph neural networks (GNNs), recurrent neural networks (RNNs), long short-term memory (LSTM), reinforcement learning (RL), transfer learning.

Selected Publications

- **M Vali**, Tom Bäckström, Arno Solin. "*DiVeQ: Differentiable Vector Quantization Using the Reparameterization Trick*". *International Conference on Learning Representations (ICLR)*, 2026 ([Link](#))
- Haishan Wang, **M Vali**, Arno Solin. "*Smol-GS: Compact Representations for Abstract 3D Gaussian Splatting*". *Submitted to Neural Information Processing Systems (NeurIPS)*, 2026 ([Link](#))
- Tom Bäckström, **M Vali**, My Nguyen, Silas Rech. "*Privacy Disclosure of Similarity Rank in Speech and Language Processing*". *IEEE Transactions on Audio, Speech and Language Processing*, 2025 ([Link](#))
- **M Vali**, Tom Bäckström. "*Unsupervised Panoptic Interpretation of Latent Spaces in GANs Using Space-Filling Vector Quantization*". *Transactions on Machine Learning Research (TMLR)*, 2025 ([Link](#))
- **M Vali**, Tom Bäckström. "*Interpretable Latent Space Using Space-Filling Curves for Phonetic Analysis in Voice Conversion*". *Interspeech Conference 2023* ([Link](#))
- **M Vali**, Tom Bäckström. "*NSVQ: Noise Substitution in Vector Quantization for Machine Learning*". *IEEE Access*, 2022 ([Link](#))

Supervision Experience

■ PhD Co-Advisor

- **Haishan Wang** Thesis: *Compressing 3D Gaussian Splatting* (2025 – Present)
- **Inshu Chauhan** Thesis: *Point Clouds Synthesis for Semantic Segmentation of Construction Scenes* (2026)

■ Master Co-Advisor

- **Terho Loikkanen** Thesis: *Explainable Analysis of Speaker Model Embeddings in DNNs* (2024)
- **Silas Rech** Thesis: *Targeted Speech Separation in Open Office Scenario Using DNNs* (2022)

■ Internships

- **Tharun Anand** Topic: *Interpreting the Latent Spaces in Diffusion Models* (2025)
- **Karahan Saritaş** Topic: *Self-Supervised Representation Learning in Vision Transformers* (2025)

Awards

- Nokia Foundation – *Doctoral research excellence scholarship* (Sep 2023)
- Walter Ahlström Foundation – *Doctoral research excellence scholarship* (Feb 2024)
- Ernst Wirtzen Foundation – *Research visit grant* (Dec 2023)
- Aalto University Science and Technology Foundation – *Research visit grant* (May 2023)

References

- **Arno Solin**, Associate Professor at Aalto University, *Postdoc Supervisor*, arno.solin@aalto.fi
- **Tom Bäckström**, Associate Professor at Aalto University, *Ph.D. Supervisor*, tom.backstrom@aalto.fi